



Crowd Sense: AI for Safer Festivals & Events

Abstract:

Big events like concerts, festivals and stadium matches need smart crowd monitoring to keep people safe and manage crowds better. This project compares two ways to analyze crowds from camera images. One uses simple texture features (LoG, entropy, HOG in HSV color) + K-means clustering to find dense crowd areas without any training data. The other uses a 3-column MCNN deep learning model trained on ShanghaiTech dataset to create density maps and count people accurately. Tests on sample images and real event photos show both methods work well together: deep learning gives smooth and precise results, especially for spread-out people, while the simple method quickly spots very crowded spots using texture only. This shows label-free methods can be powerful, and combining both could make real-time crowd safety tools even better for future events.

Keywords: Crowd Monitoring, Event Safety, Crowd Counting, Density Estimation, MCNN, Texture Features, K-means, Computer Vision, Hybrid AI

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